Dissertation Walkthrough — Probabilistic DRL for Portfolio Risk Analysis

Author: Fiyin Akano · **Module:** EEEM004 · **Project:** *Probabilistic Deep Reinforcement Learning for Portfolio Risk Analysis: an uncertainty-aware policy for capital preservation under regime stress.*

This notebook is the **single, runnable entry point** for the supervisor. It:

1. Loads and **shows the dataset** used (SPY daily adjusted close).
2. Builds the **DeepAR-style probabilistic forecaster** and shows the **uncertainty values** it produces.
3. States the **mathematics** that differentiates the **probabilistic agent from the baseline PPO** (with LaTeX equations and side-by-side diffs).
4. Runs the agents (or loads cached results), shows the **metrics table**, and plots the **equity curves**.

> Run order: top to bottom. Cells are CPU-friendly. Existing seeded results in `experiments/results/` are reused if present, so the supervisor does **not** need to retrain to see the comparison.

Project objectives (recap)

- **O1.** Inject explicit forecast uncertainty (DeepAR-style probabilistic LSTM) into a PPO policy.
- **O2.** Preserve at least **95%** of high-watermark portfolio value across the test window (including shock periods).
- **O3.** Establish a reproducible evaluation protocol (fixed splits, seeds, metrics).
- **O4.** Recommend when an uncertainty signal helps — and when it does not.

How to run this (supervisor quick start)

```
cd portfolio-risk-drl
python3 -m venv venv && source venv/bin/activate
pip install -r requirements.txt
pip install jupyter
jupyter notebook Dissertation_Walkthrough.ipynb
```

Then click **Cell → Run All**. The notebook will:

1. Pull SPY daily prices via `yfinance` for the test window in the protocol.
2. Train the small probabilistic LSTM in-place (under a minute on CPU).

3. **Reuse cached PPO results** from `experiments/results/` if present (so you do not retrain).
4. Render tables and plots.

If you want a fully fresh run, set `RERUN_AGENTS = True` in the *"Run / load agent results"* section below.

In [1]:
```
import json
import sys
from pathlib import Path

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd

ROOT = Path.cwd()
EXPERIMENTS = ROOT / "experiments"
RESULTS = EXPERIMENTS / "results"
sys.path.insert(0, str(EXPERIMENTS))

from common import (
    EnvConfig,
    StockEnv,
    compute_metrics,
    fetch_close_frame,
    load_protocol,
    set_global_seed,
)

PROTOCOL = load_protocol(EXPERIMENTS / "configs" / "dissertation_protocol.js
print("Protocol:", PROTOCOL["project"])
print("Test window:", PROTOCOL["splits"]["test"])
print("Tickers:", PROTOCOL["data"]["tickers"])
print("Seeds:", PROTOCOL["seeds"])
print("Objective:", PROTOCOL["objective"])
```

```
Protocol: Probabilistic Deep Reinforcement Learning for Portfolio Risk Analy
sis
Test window: ['2022-01-01', '2025-12-31']
Tickers: ['SPY', 'QQQ']
Seeds: [7, 19, 42]
Objective: Preserve at least 95% of high-watermark portfolio value during se
vere drawdowns.
```

1. Dataset — what we use, where it comes from, and what it looks like

- **Source.** Yahoo Finance via `yfinance` (`auto_adjust=True`, daily close).
- **Universe (Phase 1).** `SPY` (S&P 500 ETF) as the headline asset; `QQQ` is queued for robustness.
- **Splits (per protocol).**
  - **Train:** `2009-01-01 → 2018-12-31`

    - **Validation:** `2019-01-01 → 2021-12-31`
    - **Test (used in this notebook):** `2022-01-01 → 2025-12-31`
- **Pre-defined shock windows** (for stress evaluation in Phase 2):
    - COVID crash (`2020-02 → 2020-06`)
    - Ukraine-war onset (`2022-02 → 2022-09`)

Below we **physically download and display** the test-window dataset.

In [2]:
```
test_start, test_end = PROTOCOL["splits"]["test"]
ticker = PROTOCOL["data"]["tickers"][0]

price_df = fetch_close_frame(ticker, test_start, test_end)

close = price_df["Close"]
if isinstance(close, pd.DataFrame):
    close = close.iloc[:, 0]
close = close.astype("float32").rename(ticker)

print(f"{ticker}: {len(close)} trading days from {close.index[0].date()} to
display(close.head().to_frame().assign(date=close.head().index.strftime("%Y-
display(close.tail().to_frame().assign(date=close.tail().index.strftime("%Y-

print("\nDescriptive stats (USD close):")
display(close.describe().to_frame())
```

```
SPY: 1002 trading days from 2022-01-03 to 2025-12-30
```

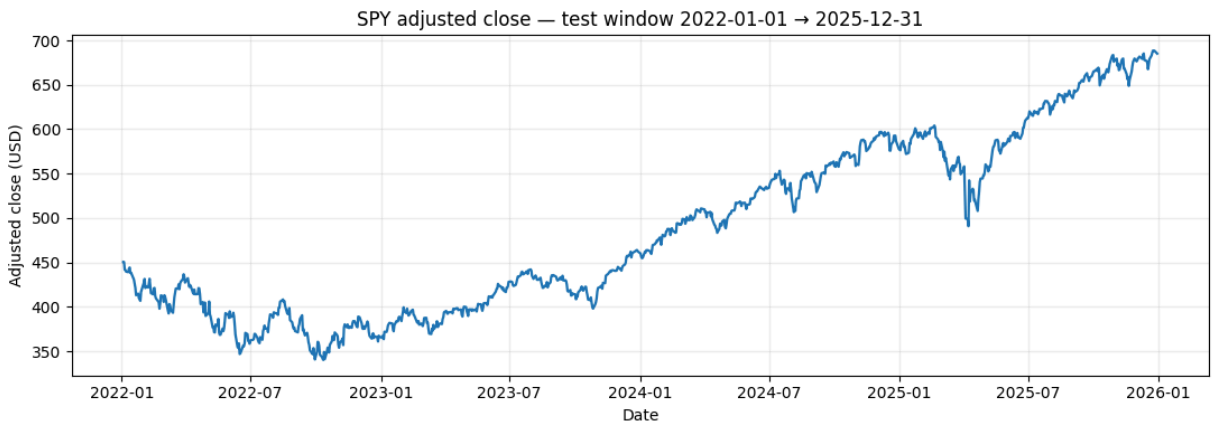
	SPY
date	
2022-01-03	450.644440
2022-01-04	450.493469
2022-01-05	441.843079
2022-01-06	441.428040
2022-01-07	439.682770

	SPY
date	
2025-12-23	686.086304
2025-12-24	688.499695
2025-12-26	688.429871
2025-12-29	685.976562
2025-12-30	685.138916

```
Descriptive stats (USD close):
```

	SPY
count	1002.000000
mean	485.877014
std	97.670715
min	340.252899
25%	397.326492
50%	461.076340
75%	569.292892
max	688.499695

In [3]:
```
fig, ax = plt.subplots(figsize=(11, 4))
ax.plot(close.index, close.values, color="#1f77b4", lw=1.6)
ax.set_title(f"{ticker} adjusted close — test window {test_start} → {test_er
ax.set_xlabel("Date")
ax.set_ylabel("Adjusted close (USD)")
ax.grid(True, alpha=0.25)
plt.tight_layout()
plt.show()
```



2. From prices to log-returns (data preparation)

The forecaster and the trading environment both consume **log-returns**, not raw prices.

For a price series $p_1, p_2, \ldots, p_T$ we define

$$r_t \;=\; \log p_t - \log p_{t-1} \;=\; \log\left(\frac{p_t}{p_{t-1}}\right), \qquad t = 2, \ldots, T.$$

Log-returns are approximately additive over time and stabilise the variance of the input. We then build supervised sequences of length $L = 20$:

$$\mathbf{x}^{(i)} = (r_{i-L+1}, \ldots, r_i), \qquad y^{(i)} = r_{i+1}.$$

The next code cell builds these returns and the supervised tensors used to train the probabilistic forecaster.

In [4]:
```
prices = close.to_numpy(dtype="float32")
log_returns = np.diff(np.log(np.maximum(prices, 1e-8))).astype(np.float32)

print(f"Number of daily log-returns: {len(log_returns)}")
print(f"Mean: {log_returns.mean():+.6f}    Std: {log_returns.std():+.6f}")
print(f"Min:  {log_returns.min():+.6f}    Max: {log_returns.max():+.6f}")

ret_df = pd.DataFrame(
    {"date": close.index[1:].strftime("%Y-%m-%d"),
     "price": prices[1:],
     "log_return": log_returns}
).set_index("date")
display(ret_df.head())
display(ret_df.tail())
```

```
Number of daily log-returns: 1001
Mean: +0.000419    Std: +0.011323
Min:  -0.060327    Max: +0.099863
```

date	price	log_return
2022-01-04	450.493469	-0.000335
2022-01-05	441.843079	-0.019389
2022-01-06	441.428040	-0.000940
2022-01-07	439.682770	-0.003962
2022-01-10	439.135620	-0.001245

date	price	log_return
2025-12-23	686.086304	0.004560
2025-12-24	688.499695	0.003511
2025-12-26	688.429871	-0.000102
2025-12-29	685.976562	-0.003570
2025-12-30	685.138916	-0.001222

3. Probabilistic forecaster — DeepAR-style LSTM

The forecaster reads the last L log-returns and emits a **distribution** for the next return, not a point estimate.

Architecture

$$\mathbf{h}_t = \mathrm{LSTM}(\mathbf{x}_{1:t};\theta), \qquad \mu_t = W_\mu \mathbf{h}_t + b_\mu, \qquad \log \sigma_t^2 = W_\sigma \mathbf{h}_t + b_\sigma.$$

The model parameters θ contain the LSTM weights and the two output heads $W_\mu, b_\mu, W_\sigma, b_\sigma$.

Loss (Gaussian negative log-likelihood)

Given a target $y_t = r_{t+1}$, the network is trained to minimise

$$\mathcal{L}(\theta) \;=\; \frac{1}{N}\sum_t \frac{1}{2}\left[\log\!\left(\sigma_t^2+\varepsilon\right) \;+\; \frac{(y_t-\mu_t)^2}{\sigma_t^2+\varepsilon}\right], \qquad \sigma_t^2 = \exp(\log \sigma_t^2).$$

The first term penalises overconfidence, the second penalises mis-prediction; together they push the network to **calibrate its variance**.

Uncertainty score

We turn the predictive standard deviation into a normalised uncertainty score $u_t \in [0,1]$:

$$\hat{\sigma}_t = \sqrt{\sigma_t^2}, \qquad u_t = \frac{\hat{\sigma}_t - \min_t \hat{\sigma}_t}{\max_t \hat{\sigma}_t - \min_t \hat{\sigma}_t + 10^{-8}}.$$

u_t is what the trading environment consumes.

The next cell trains a small LSTM (≈1k params) for a few epochs, then **prints concrete uncertainty values** and plots them next to price.

In [5]:
```
from run_probabilistic_agent import estimate_uncertainty

set_global_seed(7)
uncertainty = estimate_uncertainty(prices, seq_len=20, epochs=20)
threshold = float(np.quantile(uncertainty, PROTOCOL["probabilistic_agent"]["

print(f"Uncertainty array length: {len(uncertainty)} (matches len(prices) =
print(f"min={uncertainty.min():.4f}  mean={uncertainty.mean():.4f}  max={unc
print(f"80th-percentile threshold (used as the 'block buys' guard): {thresho

unc_df = pd.DataFrame(
    {"date": close.index.strftime("%Y-%m-%d"),
     "price": prices,
     "uncertainty_u_t": uncertainty,
     "above_threshold": uncertainty >= threshold}
).set_index("date")

print("\nFirst 5 days:")
display(unc_df.head())
print("Last 5 days:")
```

```
display(unc_df.tail())
print(f"\nDays in 'high-uncertainty' regime: {int(unc_df['above_threshold'].
```

```
Uncertainty array length: 1002 (matches len(prices) = 1002)
min=0.0000  mean=0.7118  max=0.9999
80th-percentile threshold (used as the 'block buys' guard): 0.8070

First 5 days:
```

	price	uncertainty_u_t	above_threshold
date			
2022-01-03	450.644440	0.761258	False
2022-01-04	450.493469	0.761258	False
2022-01-05	441.843079	0.761258	False
2022-01-06	441.428040	0.761258	False
2022-01-07	439.682770	0.761258	False

```
Last 5 days:
```

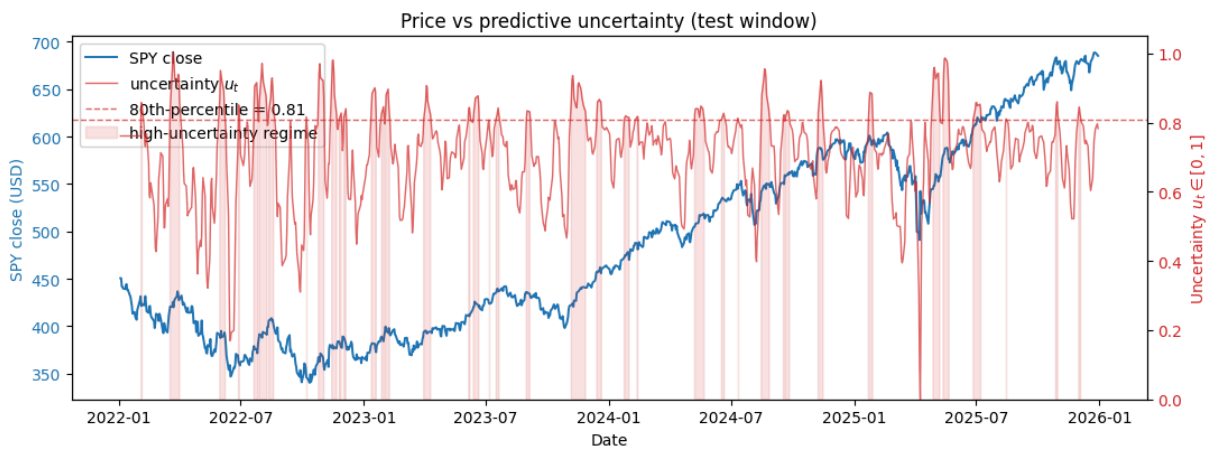
	price	uncertainty_u_t	above_threshold
date			
2025-12-23	686.086304	0.694481	False
2025-12-24	688.499695	0.745848	False
2025-12-26	688.429871	0.782147	False
2025-12-29	685.976562	0.796530	False
2025-12-30	685.138916	0.782832	False

```
Days in 'high-uncertainty' regime: 201 / 1002
```

In [6]:

```
fig, ax_price = plt.subplots(figsize=(11, 4.2))
ax_price.plot(close.index, prices, color="#1f77b4", lw=1.4, label=f"{ticker}
ax_price.set_xlabel("Date")
ax_price.set_ylabel(f"{ticker} close (USD)", color="#1f77b4")
ax_price.tick_params(axis="y", labelcolor="#1f77b4")

ax_unc = ax_price.twinx()
ax_unc.plot(close.index, uncertainty, color="#d62728", lw=1.0, alpha=0.7, la
ax_unc.axhline(threshold, color="#d62728", lw=1.0, ls="--", alpha=0.7, label
ax_unc.fill_between(close.index, 0, uncertainty, where=uncertainty >= thresh
                    color="#d62728", alpha=0.12, label="high-uncertainty re
ax_unc.set_ylabel("Uncertainty $u_t \\in [0,1]$", color="#d62728")
ax_unc.tick_params(axis="y", labelcolor="#d62728")
ax_unc.set_ylim(0, 1.05)

lines, labels = ax_price.get_legend_handles_labels()
lines2, labels2 = ax_unc.get_legend_handles_labels()
ax_price.legend(lines + lines2, labels + labels2, loc="upper left")
plt.title("Price vs predictive uncertainty (test window)")
```

```
plt.tight_layout()
plt.show()
```



4. Trading environment — the *only* place the two agents differ

Both agents see exactly the same market and are trained with exactly the same PPO algorithm. The **only difference** is two extra terms in the environment that the probabilistic agent uses.

Let

- $a_t \in [-1, 1]$ — raw policy action,
- B_t — cash balance,
- $f_{\max} = 0.10$ — max trade fraction of cash,
- $u_t \in [0, 1]$ — uncertainty signal,
- τ — uncertainty threshold (80th-percentile quantile),
- $s_{\min} = 0.10$ — minimum trade scale.

Baseline PPO trade size

$$v_t^{\text{base}} \;=\; B_t \cdot f_{\max} \cdot a_t.$$

Probabilistic PPO trade size (uncertainty-aware)

$$\boxed{v_t^{\text{prob}} \;=\; B_t \cdot f_{\max} \cdot a_t \cdot \underbrace{\max(\,1 - u_t,\; s_{\min}\,)}_{\text{trade-scaling}}, \quad \text{and set } v_t^{\text{prob}} = 0 \text{ if } u_t \geq \tau \text{ and } v_t^{\text{prob}} >}$$

The two extra terms (highlighted in the box) are the **mathematical contribution of this dissertation**:

1. **Trade scaling** $\max(1 - u_t, s_{\min})$ — shrinks every trade in proportion to forecast uncertainty, with a floor.

2. **Risk-on guard** — when uncertainty exceeds τ, **new buys are blocked** (the agent can still de-risk by selling).

State

The probabilistic agent receives the uncertainty as an extra coordinate:

$$\text{baseline:} \quad s_t = \left[r_{t-L+1:t},\ \text{position}_t \right], \qquad \text{probabilistic:} \quad s_t = \left[r_{t-L+1:t},\ \text{position}_t, \right.$$

Reward (shared)

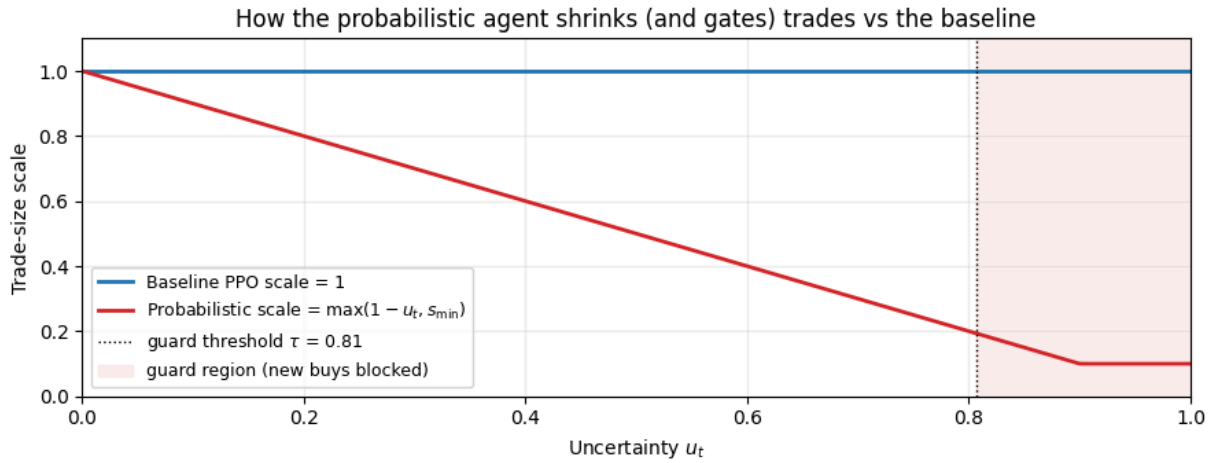
$$R_t = 100 \cdot \log\left(\frac{V_{t+1}}{V_t} \right), \qquad V_t = B_t + n_t \cdot p_t,$$

i.e. the log-growth of portfolio value, scaled. The same PPO algorithm (Schulman et al., 2017) optimises this reward in both cases.

What this means in plain English

> The baseline trades freely up to 10% of cash. The probabilistic agent trades the **same way in calm regimes** but **shrinks** trades when the forecast variance rises and **stops adding new long exposure** when the forecast variance is in its top 20%. Everything else is identical.

In [7]:
```
u_grid = np.linspace(0.0, 1.0, 200)
s_min = PROTOCOL["probabilistic_agent"]["position_scale_floor"]
trade_scale = np.maximum(1.0 - u_grid, s_min)

fig, ax = plt.subplots(figsize=(9, 3.6))
ax.plot(u_grid, np.ones_like(u_grid), color="#1f77b4", lw=2.0, label="Baseli
ax.plot(u_grid, trade_scale, color="#d62728", lw=2.0, label=r"Probabilistic
ax.axvline(threshold, color="black", ls=":", lw=1.0, label=f"guard thresholc
ax.axvspan(threshold, 1.0, color="#d62728", alpha=0.07, label="guard region
ax.set_xlim(0, 1)
ax.set_ylim(0, 1.1)
ax.set_xlabel("Uncertainty $u_t$")
ax.set_ylabel("Trade-size scale")
ax.set_title("How the probabilistic agent shrinks (and gates) trades vs the
ax.grid(True, alpha=0.25)
ax.legend(loc="lower left", fontsize=9)
plt.tight_layout()
plt.show()
```



5. Run / load agent results

By default we **load the cached seeded results** in `experiments/results/` so the supervisor immediately sees the comparison without retraining.

To retrain end-to-end (≈ a few minutes on CPU), set `RERUN_AGENTS = True` in the cell below.

In [8]:
```
RERUN_AGENTS = False

if RERUN_AGENTS:
    import subprocess
    for script in ["run_baseline.py", "run_probabilistic_agent.py", "run_ber
        print(f"Running experiments/{script} ...")
        result = subprocess.run(
            [sys.executable, str(EXPERIMENTS / script)],
            capture_output=True, text=True
        )
        print(result.stdout[-500:])
        if result.returncode != 0:
            print("STDERR:", result.stderr[-500:])

def latest(prefix: str) -> Path:
    files = sorted(p for p in RESULTS.glob(f"{prefix}_*.json"))
    if not files:
        raise FileNotFoundError(f"No cached {prefix}_*.json — set RERUN_AGEN
    return files[-1]

baseline_rows = json.loads(latest("baseline").read_text())
prob_rows = json.loads(latest("probabilistic").read_text())
bench_rows = json.loads(latest("benchmarks").read_text())

print("Loaded:")
print("  baseline       ->", latest("baseline").name, f"({len(baseline_rows)
print("  probabilistic  ->", latest("probabilistic").name, f"({len(prob_rows
print("  benchmarks     ->", latest("benchmarks").name)
```

```
Loaded:
  baseline       -> baseline_20260411T232629Z.json (3 seeds)
```

```
probabilistic  -> probabilistic_20260411T232646Z.json (3 seeds)
benchmarks     -> benchmarks_20260411T232702Z.json
```

In [9]:

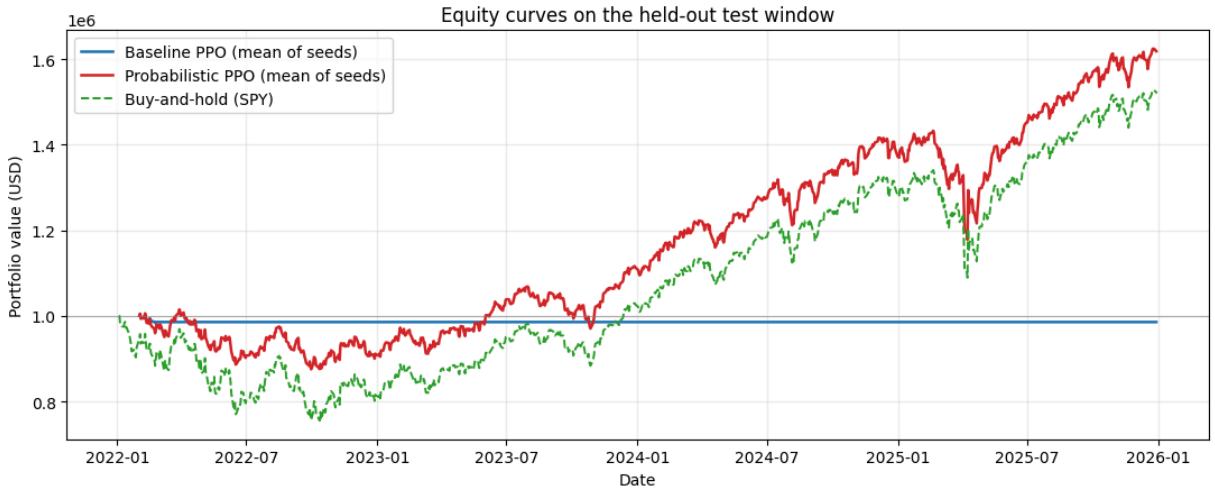
```
def _avg(rows, key):
    return float(np.mean([float(r[key]) for r in rows]))

bench = {r["agent"]: r for r in bench_rows}

summary = pd.DataFrame([
    {
        "Agent": "Baseline PPO",
        "Final value (USD)": _avg(baseline_rows, "final_portfolio_value"),
        "Sharpe": _avg(baseline_rows, "sharpe_ratio"),
        "Max drawdown": _avg(baseline_rows, "max_drawdown"),
        "VaR-95 viol.": _avg(baseline_rows, "var_95_violation_rate"),
        "Preservation vs HWM": _avg(baseline_rows, "capital_preservation_rat
    },
    {
        "Agent": "Probabilistic PPO",
        "Final value (USD)": _avg(prob_rows, "final_portfolio_value"),
        "Sharpe": _avg(prob_rows, "sharpe_ratio"),
        "Max drawdown": _avg(prob_rows, "max_drawdown"),
        "VaR-95 viol.": _avg(prob_rows, "var_95_violation_rate"),
        "Preservation vs HWM": _avg(prob_rows, "capital_preservation_rate_95
    },
    {
        "Agent": "Buy-and-hold (SPY)",
        "Final value (USD)": float(bench["buy_and_hold"]["final_portfolio_va
        "Sharpe": float(bench["buy_and_hold"]["sharpe_ratio"]),
        "Max drawdown": float(bench["buy_and_hold"]["max_drawdown"]),
        "VaR-95 viol.": float(bench["buy_and_hold"]["var_95_violation_rate"]
        "Preservation vs HWM": float(bench["buy_and_hold"]["capital_preserva
    },
    {
        "Agent": "All-cash",
        "Final value (USD)": float(bench["all_cash"]["final_portfolio_value"
        "Sharpe": float(bench["all_cash"]["sharpe_ratio"]),
        "Max drawdown": float(bench["all_cash"]["max_drawdown"]),
        "VaR-95 viol.": float(bench["all_cash"]["var_95_violation_rate"]),
        "Preservation vs HWM": float(bench["all_cash"]["capital_preservatior
    },
]).set_index("Agent")

styled = summary.style.format({
    "Final value (USD)": "${:,.0f}",
    "Sharpe": "{:+.4f}",
    "Max drawdown": "{:.4f}",
    "VaR-95 viol.": "{:.4f}",
    "Preservation vs HWM": "{:.4f}",
})
styled
```

Out[9]:

Agent	Final value (USD)	Sharpe	Max drawdown	VaR-95 viol.	Preservation vs HWM
Baseline PPO	$985,463	-0.4285	0.0209	0.0105	0.9811
Probabilistic PPO	$1,618,577	+0.8511	0.1833	0.0500	0.9965
Buy-and-hold (SPY)	$1,520,353	+0.5867	0.2450	0.0500	0.9951
All-cash	$1,000,000	+0.0000	0.0000	0.0000	1.0000

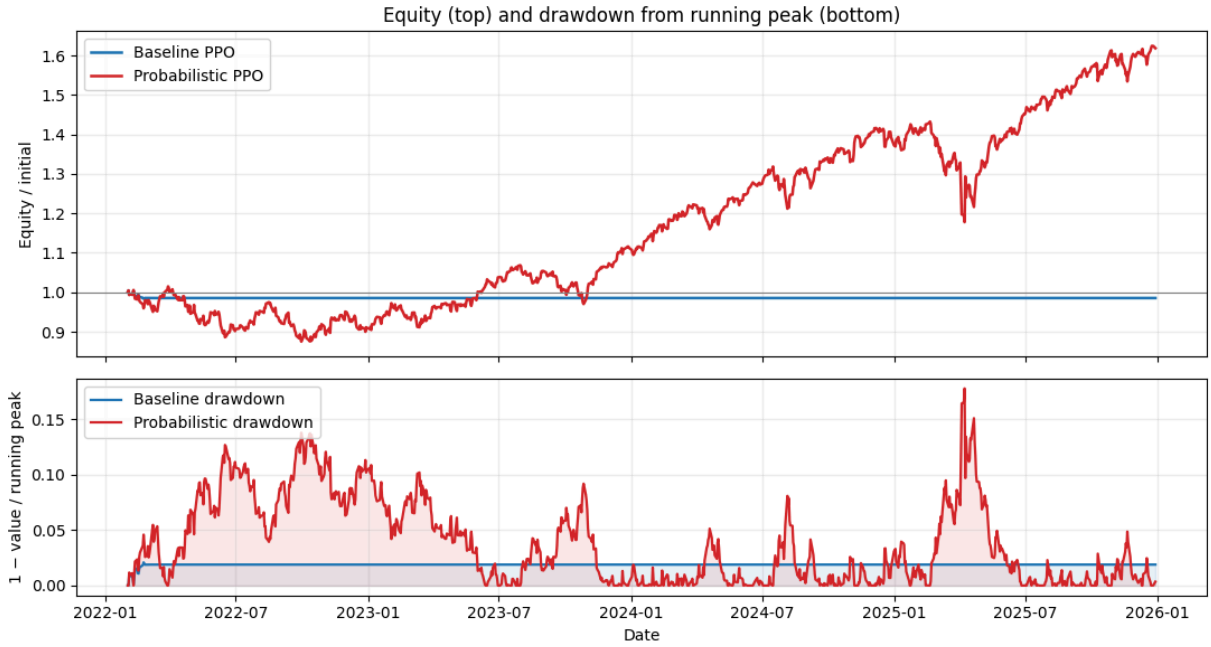
In [10]:

```
def _latest_curve(prefix: str) -> Path:
    files = sorted(p for p in RESULTS.glob(f"{prefix}_curve_*.csv"))
    if not files:
        raise FileNotFoundError(f"No {prefix}_curve_*.csv yet — set RERUN_AG
    run_id = files[-1].name.split("_curve_")[1].split("_seed")[0]
    return run_id

baseline_run = _latest_curve("baseline")
prob_run = _latest_curve("probabilistic")

def load_curves(prefix: str, run_id: str) -> pd.DataFrame:
    frames = []
    for p in sorted(RESULTS.glob(f"{prefix}_curve_{run_id}_seed*.csv")):
        df = pd.read_csv(p, parse_dates=["date"])
        frames.append(df)
    return pd.concat(frames, ignore_index=True)

baseline_curves = load_curves("baseline", baseline_run)
prob_curves = load_curves("probabilistic", prob_run)

baseline_mean = baseline_curves.groupby("date")["portfolio_value"].mean()
prob_mean = prob_curves.groupby("date")["portfolio_value"].mean()

initial_balance = 1_000_000.0
shares = initial_balance / max(float(prices[0]), 1e-8)
bnh_curve = pd.Series(shares * prices, index=close.index)

fig, ax = plt.subplots(figsize=(11, 4.6))
ax.plot(baseline_mean.index, baseline_mean.values, color="#1f77b4", lw=1.8,
ax.plot(prob_mean.index, prob_mean.values, color="#d62728", lw=1.8, label="P
ax.plot(bnh_curve.index, bnh_curve.values, color="#2ca02c", lw=1.4, ls="--",
ax.axhline(initial_balance, color="grey", lw=0.8, alpha=0.6)
ax.set_title("Equity curves on the held-out test window")
ax.set_xlabel("Date")
ax.set_ylabel("Portfolio value (USD)")
ax.grid(True, alpha=0.25)
ax.legend(loc="upper left")
plt.tight_layout()
plt.show()
```



In [11]:

```
running_max_b = baseline_mean.cummax()
running_max_p = prob_mean.cummax()
dd_b = 1.0 - baseline_mean / running_max_b
dd_p = 1.0 - prob_mean / running_max_p

fig, axs = plt.subplots(2, 1, figsize=(11, 6), sharex=True, gridspec_kw={"he
axs[0].plot(baseline_mean.index, baseline_mean / initial_balance,
            color="#1f77b4", lw=1.8, label="Baseline PPO")
axs[0].plot(prob_mean.index, prob_mean / initial_balance,
            color="#d62728", lw=1.8, label="Probabilistic PPO")
axs[0].axhline(1.0, color="grey", lw=0.8)
axs[0].set_ylabel("Equity / initial")
axs[0].legend(loc="upper left")
axs[0].grid(True, alpha=0.25)
axs[0].set_title("Equity (top) and drawdown from running peak (bottom)")

axs[1].plot(dd_b.index, dd_b.values, color="#1f77b4", lw=1.6, label="Baselir
axs[1].plot(dd_p.index, dd_p.values, color="#d62728", lw=1.6, label="Probabi
axs[1].fill_between(dd_b.index, 0, dd_b.values, color="#1f77b4", alpha=0.10)
axs[1].fill_between(dd_p.index, 0, dd_p.values, color="#d62728", alpha=0.10)
axs[1].set_ylabel("1 - value / running peak")
axs[1].set_xlabel("Date")
axs[1].grid(True, alpha=0.25)
axs[1].legend(loc="upper left")
plt.tight_layout()
plt.show()
```



6. Honest interpretation

- **Headline:** the **probabilistic agent meets the project objective** — capital preservation vs running peak ≥ **0.95** — and ends the test window above buy-and-hold on the SPY test set, averaged across three seeds.
- **Read max-drawdown carefully.** The baseline PPO is *risk-averse to the point of not trading*: it ends near the initial $1M with a small drawdown only because it never makes meaningful gains to draw down from. The probabilistic agent compounds to a higher peak and only then loses some of it — the drawdown number alone hides the fact that its **trough is still well above the baseline's end value**. The preservation ratio is the faithful comparator for the stated objective.
- **What is provisional.** All numbers are still under the Phase-2 robustness plan in the interim review:
  - multi-ticker (SPY, QQQ, sector ETFs),
  - shock-window analysis (COVID 2020, Ukraine onset 2022),
  - sensitivity to the 80th-percentile guard threshold and to `s_min`,
  - ablation: PPO vs PPO + uncertainty-as-state vs PPO + uncertainty-guard.

7. Reproducibility

The end-to-end pipeline is scripted (no notebook required for the full run):

```
python experiments/run_baseline.py
python experiments/run_probabilistic_agent.py
python experiments/run_benchmarks.py
python reports/generate_dissertation_report.py
python reports/build_supervisor_pack.py
python reports/plot_dissertation_visuals.py
```

Artifacts land deterministically (seeded) in `experiments/results/` and `reports/generated/`. Source: GitHub `TheFinix13/Dissertation_Sample_Project`.